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ENERGY CONVERSION-I

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LECTURE-07

Text Book: Electric Machinery Fundamentals

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5th Edition

Chapter 6

Induction Motors



Inspiring Excellence

Chapter 6: Induction Motors

- Construction of Induction Motors
- Basic Induction Motor Concepts
- Equivalent circuit of an induction motor
- Power and Torque in induction motor
- Torque-speed characteristics
- Starting Induction Motors
- Speed Control of Induction Motors
- Single phase induction motors

Equivalent Circuit

- The induction motor is similar to the transformer with the exception that its secondary windings are free to rotate.
- As we noticed in the transformer, it is easier if we can combine these two circuits in one circuit but there are some difficulties

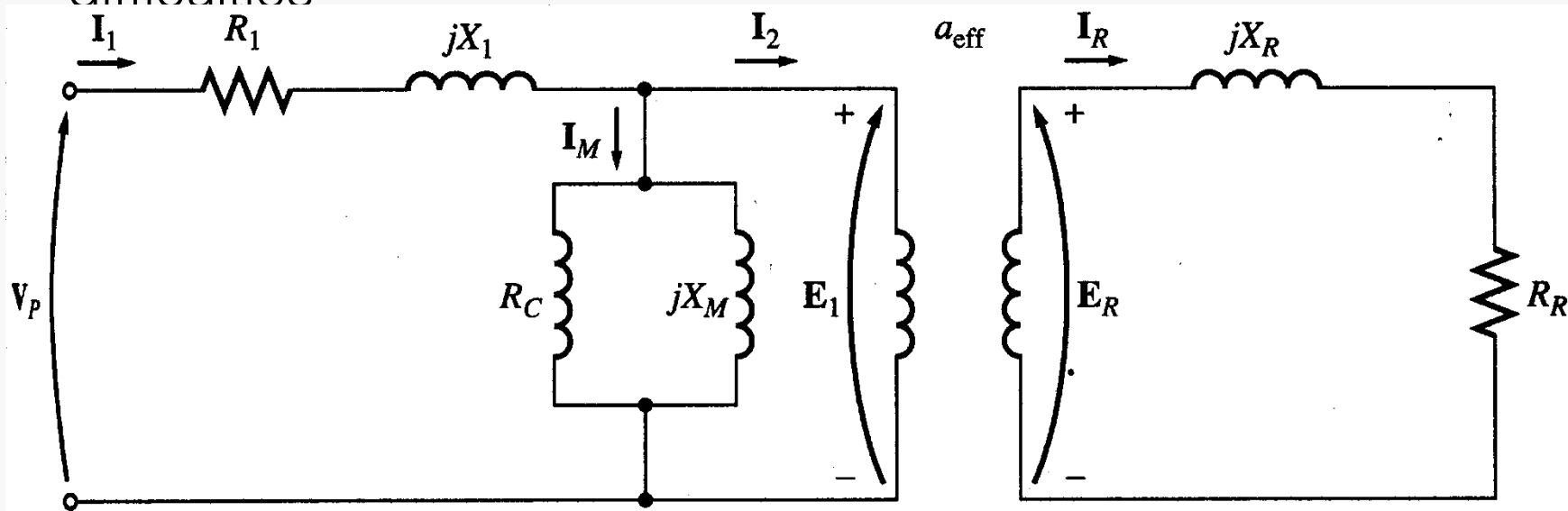


FIGURE 6-7

The transformer model of an induction motor, with rotor and stator connected by an ideal transformer of turns ratio a_{eff}

The transformer model of an induction motor

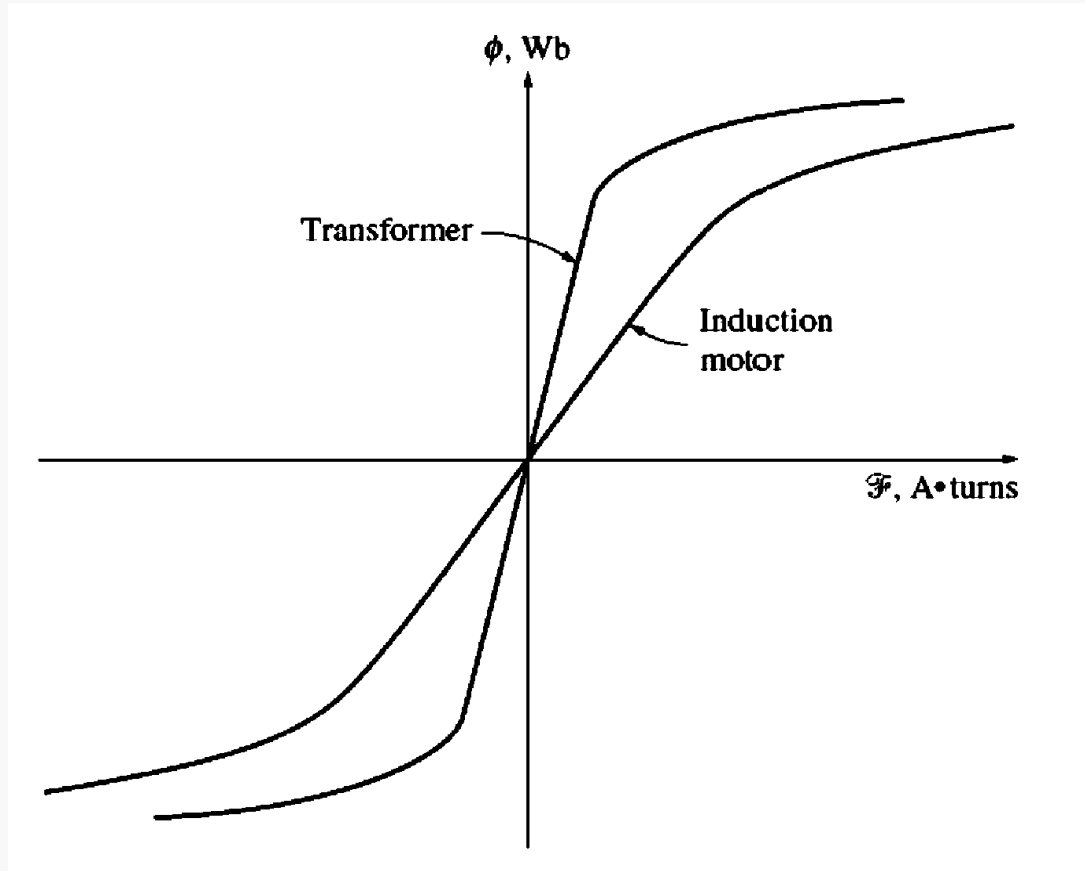


FIGURE 6-8

The magnetization curve of an induction motor compared to that of a transformer.

The magnetization curve

- Notice- the slope of the induction motor's mmf-flux curve is much shallower than the curve of a good transformer.
- Reason-there must be an air gap in an induction motor, which greatly increases the reluctance of the flux path and therefore reduces the coupling between primary and secondary windings.
- The higher reluctance caused by the air gap means that a higher magnetizing current is required to obtain a given flux level.
- Therefore, the magnetizing reactance X_M in the equivalent circuit will have a much smaller value (or the susceptance B_M will have a much larger value) than it would in an ordinary transformer.

Rotor Circuit Model

- When the rotor is locked (or blocked), i.e. $s = 1$, the largest voltage and rotor frequency are induced in the rotor.
- On the other side, if the rotor rotates at synchronous speed, i.e. $s = 0$, the induced voltage and frequency in the rotor will be equal to zero, .

$$E_R = sE_{R0}$$

Where E_{R0} is the largest value of the rotor's induced voltage obtained at $s = 1$ (locked rotor)

Rotor Circuit Model Contd.

- The same is true for the frequency, i.e.

$$f_{re} = s f_{se}$$

- So, as the frequency of the induced voltage in the rotor changes, the reactance of the rotor circuit also changes

Where X_{r0} is the rotor reactance at the supply frequency (at blocked rotor)

$$X_R = \omega_{re} L_R = 2\pi f_{re} L_R$$

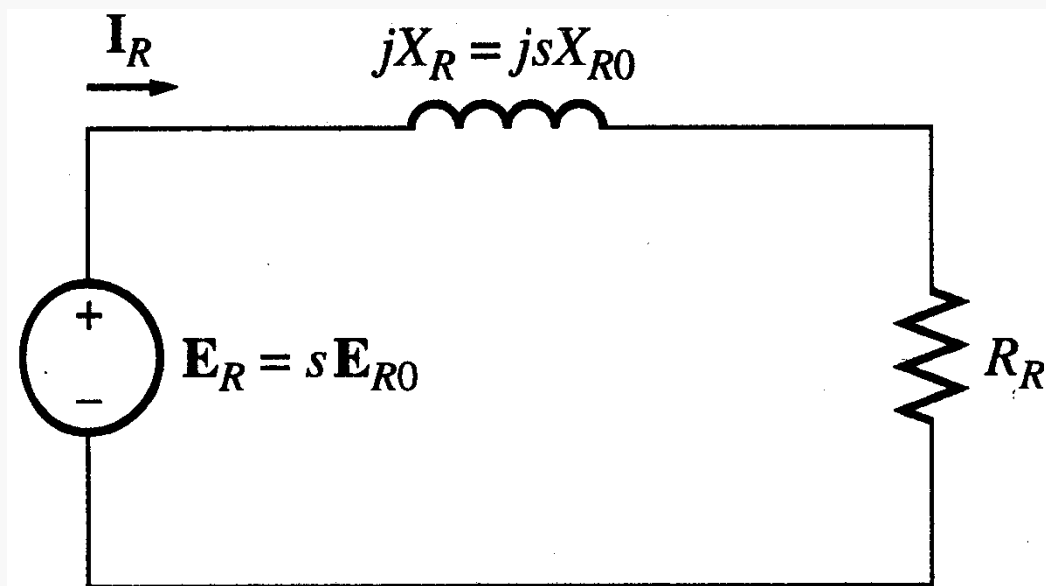
$$\begin{aligned} X_R &= 2\pi s f_{se} L_R \\ &= s(2\pi f_{se} L_R) \\ &= sX_{R0} \end{aligned}$$

Rotor Circuit Model Contd.

- Then, we can draw the rotor equivalent circuit as follows

FIGURE 6-9

The rotor circuit model of an induction motor.

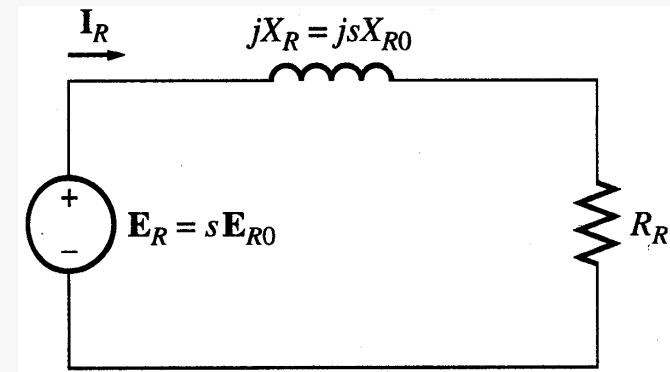


Where E_R is the induced voltage in the rotor and R_R is the rotor resistance

Rotor Circuit Model Contd.

- Now we can calculate the rotor current as

$$I_R = \frac{E_R}{(R_R + jX_R)} = \frac{sE_{R0}}{(R_R + jsX_{R0})}$$



- Dividing both the numerator and denominator by s so nothing changes we get

$$I_R = \frac{E_{R0}}{R_R/s + jX_{R0}}$$

Where E_{R0} is the induced voltage and X_{R0} is the rotor reactance at blocked rotor condition ($s = 1$)

Rotor Circuit Model Contd.

- Now we can have the rotor equivalent circuit

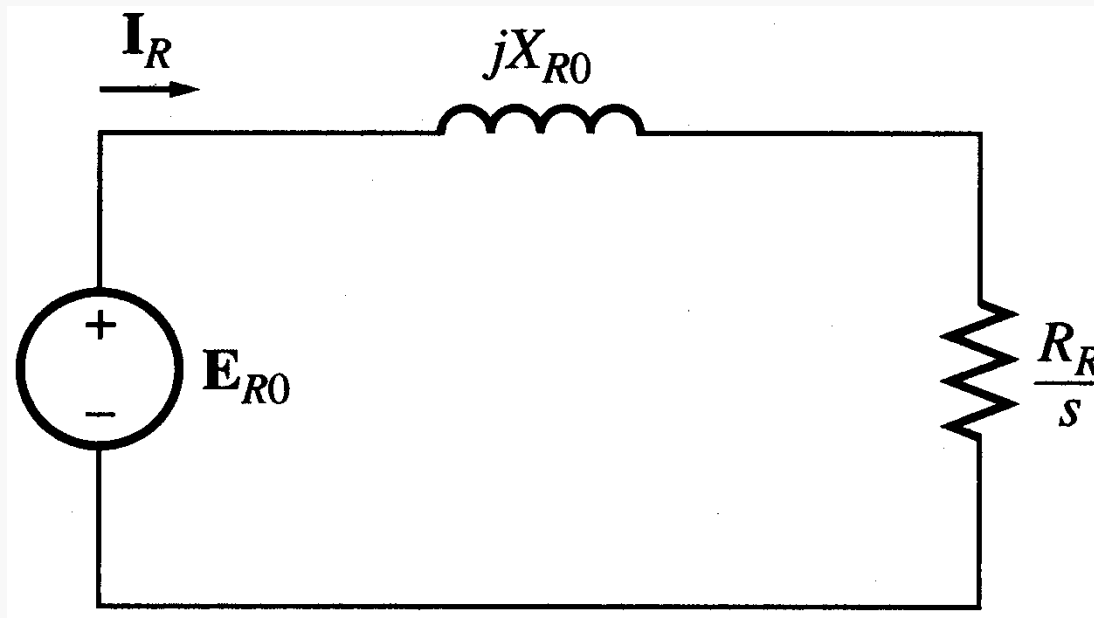


FIGURE 6-10

The rotor circuit model with all the frequency (slip) effects concentrated in resistor R_R

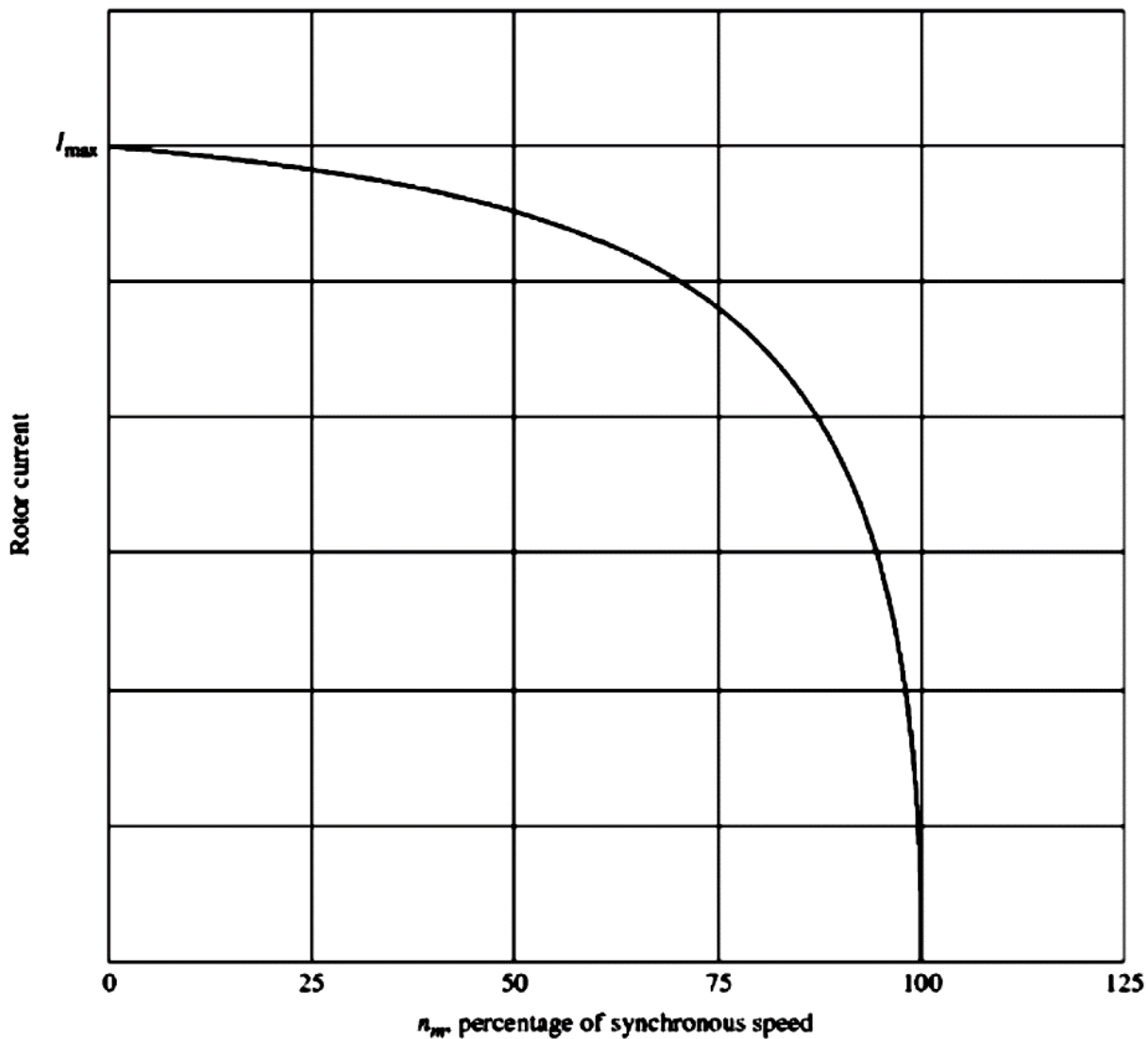


FIGURE 6-11
Rotor current as a function of rotor speed.

The Final Equivalent Circuit

- Now as we managed to solve the induced voltage and different frequency problems, we can combine the stator and rotor circuits in one equivalent circuit

Where

Exactly the same sort of transformation can be done for the induction motor's rotor circuit. If the effective turns ratio of an induction motor is a_{eff} , then the transformed rotor voltage becomes

$$\mathbf{E}_1 = \mathbf{E}'_R = a_{\text{eff}} \mathbf{E}_{R0} \quad (6-18)$$

the rotor current becomes

$$\mathbf{I}_2 = \frac{\mathbf{I}_R}{a_{\text{eff}}} \quad (6-19)$$

and the rotor impedance becomes

$$\mathbf{Z}_2 = a_{\text{eff}}^2 \left(\frac{R_R}{s} + jX_{R0} \right) \quad (6-20)$$

If we now make the following definitions:

$$R_2 = a_{\text{eff}}^2 R_R \quad (6-21)$$

$$X_2 = a_{\text{eff}}^2 X_{R0} \quad (6-22)$$

then the final per-phase equivalent circuit of the induction motor is as shown in Figure 6-12.

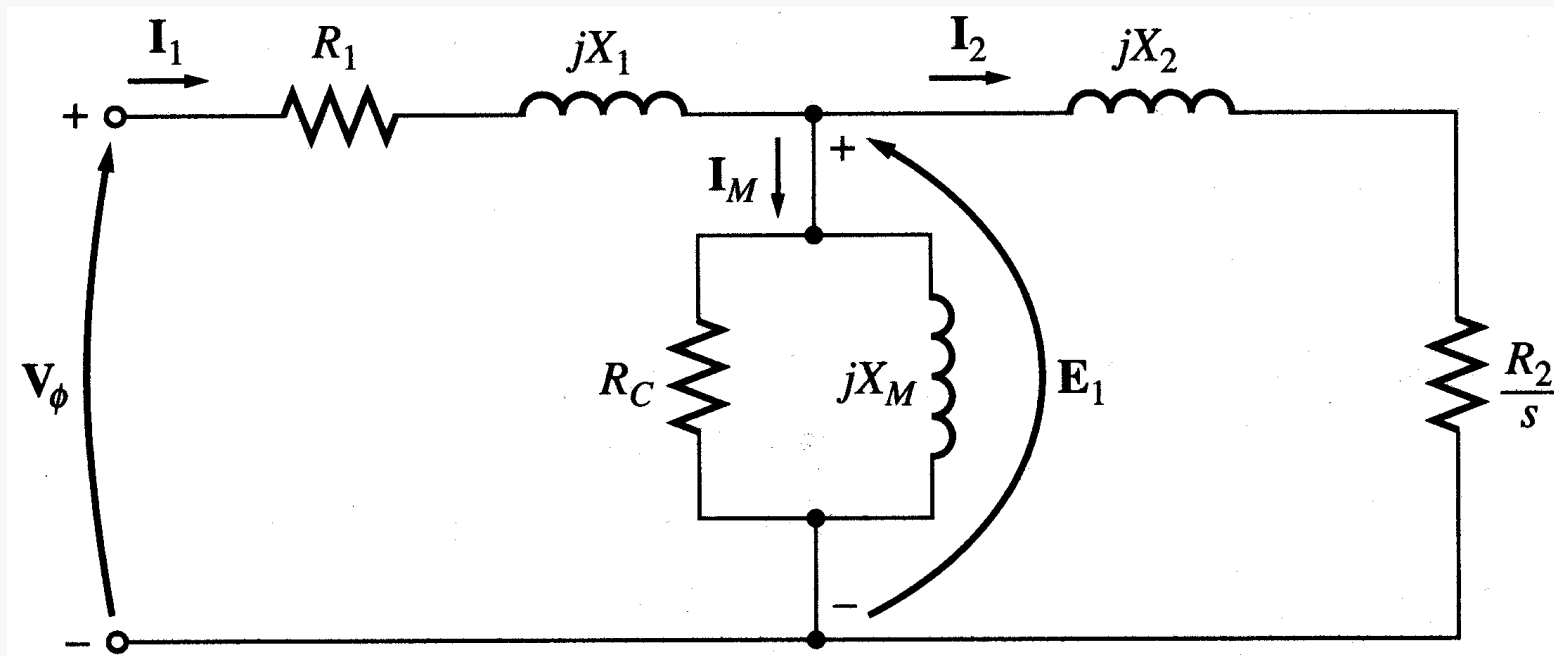


FIGURE 6-12

The per-phase equivalent circuit of an induction motor.

Power losses in Induction machines

- Copper losses
 - *Copper loss in the stator (P_{SCL}) = $I_1^2 R_1$*
 - *Copper loss in the rotor (P_{RCL}) = $I_2^2 R_2$*
- Core loss (P_{core})
- Mechanical power loss due to friction and windage
- How this power flow in the motor?

Power flow in induction motor

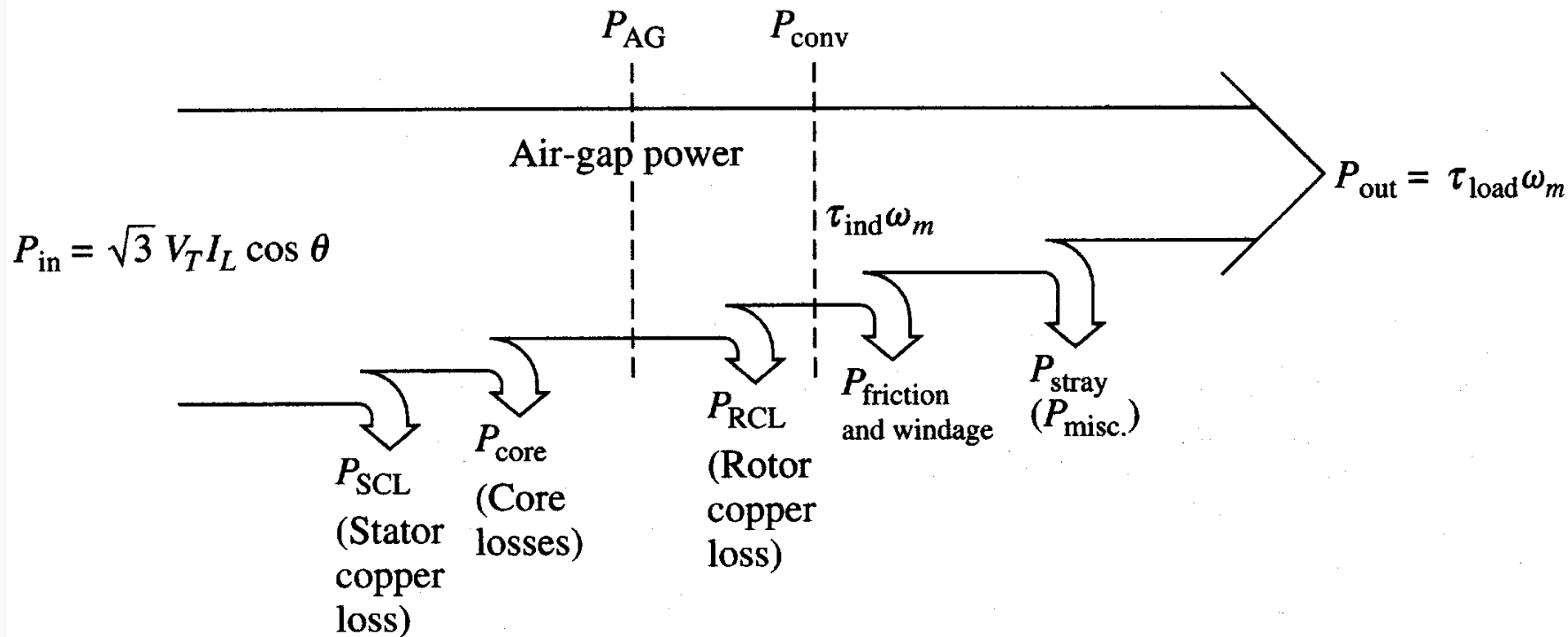


FIGURE 6-13

The power-flow diagram of an induction motor.

Example 6-2

A 480-V, 60 Hz, 50-hp, three phase induction motor is drawing 60A at 0.85 PF lagging. The stator copper losses are 2 kW, and the rotor copper losses are 700 W. The friction and windage losses are 600 W, the core losses are 1800 W, and the stray losses are negligible. Find the following quantities:

1. *The air-gap power P_{AG} .*
2. *The power converted P_{conv} .*
3. *The output power P_{out} .*
4. *The efficiency of the motor.*

Solution

$$\begin{aligned} 1. \quad P_{in} &= \sqrt{3} V_L I_L \cos \theta \\ &= \sqrt{3} \times 480 \times 60 \times 0.85 = 42.4 \text{ kW} \end{aligned}$$

$$\begin{aligned} P_{AG} &= P_{in} - P_{SCL} - P_{core} \\ &= 42.4 - 2 - 1.8 = 38.6 \text{ kW} \end{aligned}$$

$$\begin{aligned} 2. \quad P_{conv} &= P_{AG} - P_{RCL} \\ &= 38.6 - \frac{700}{1000} = 37.9 \text{ kW} \end{aligned}$$

$$\begin{aligned} 3. \quad P_{out} &= P_{conv} - P_{F\&W} \\ &= 37.9 - \frac{600}{1000} = 37.3 \text{ kW} \end{aligned}$$

Solution

$$P_{out} = \frac{37.3}{0.746} = 50 \text{ hp}$$

$$\begin{aligned} 4. \quad \eta &= \frac{P_{out}}{P_{in}} \times 100\% \\ &= \frac{37.3}{42.4} \times 100 = 88\% \end{aligned}$$

Power and Torque relations

$$P_{in} = \sqrt{3} V_L I_L \cos \theta = 3 V_{ph} I_{ph} \cos \theta$$

$$P_{SCL} = 3 I_1^2 R_1$$

$$P_{AG} = P_{in} - (P_{SCL} + P_{core}) = P_{conv} + P_{RCL} = 3 I_2^2 \frac{R_2}{s} = \frac{P_{RCL}}{s}$$

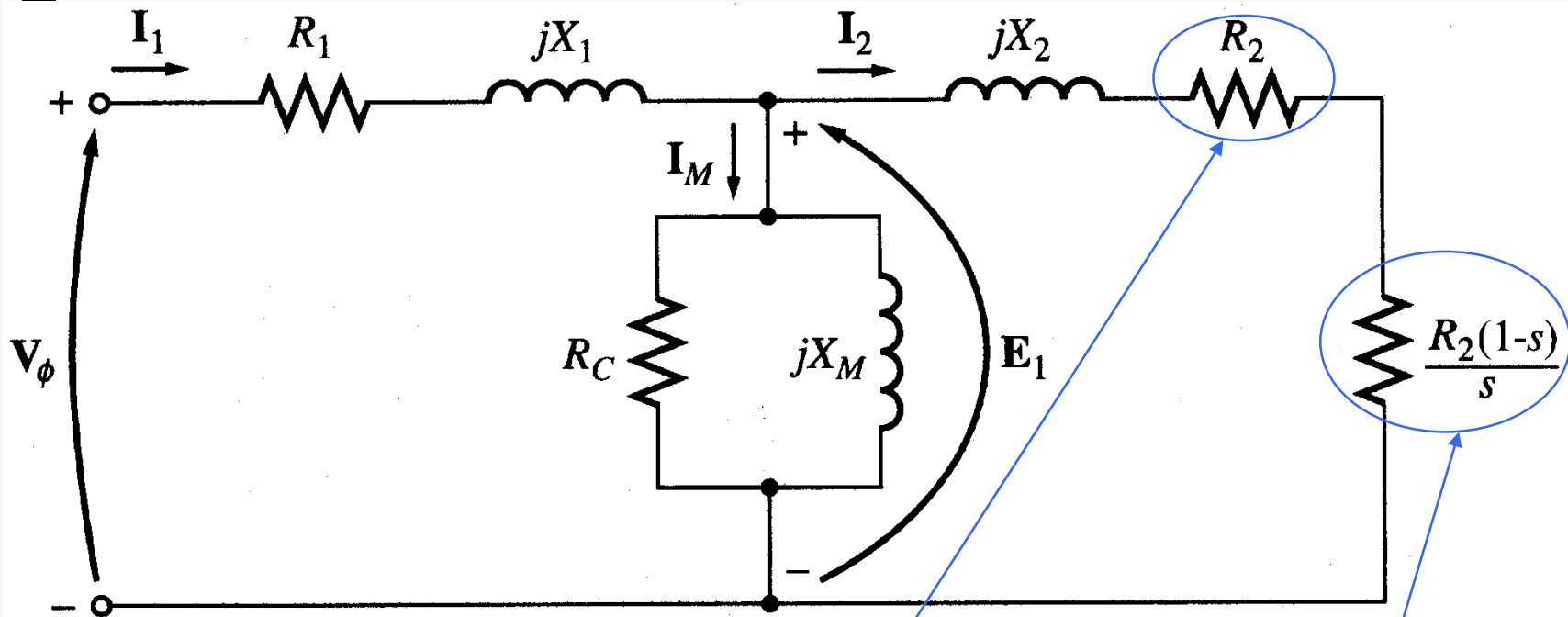
$$P_{RCL} = 3 I_2^2 R_2$$

$$P_{conv} = P_{AG} - P_{RCL} = 3 I_2^2 \frac{R_2 (1-s)}{s} = \frac{P_{RCL} (1-s)}{s}$$

$$P_{conv} = (1-s) P_{AG}$$

$$P_{out} = P_{conv} - (P_{f+w} + P_{stray}) \quad \tau_{ind} = \frac{P_{conv}}{\omega_m} = \frac{(1-s) P_{AG}}{(1-s) \omega_s}$$

Separating the Rotor Copper Losses and the Power Converted in an Induction Motor's Equivalent Circuit



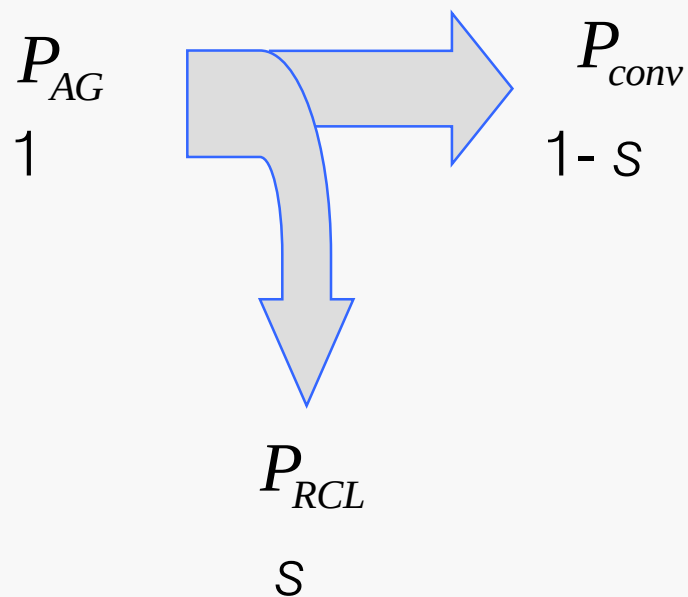
$$R_{\text{conv}} = \frac{R_2}{s} - R_2 = R_2 \left(\frac{1}{s} - 1 \right)$$

$$R_{\text{conv}} = R_2 \left(\frac{1-s}{s} \right)$$

Actual rotor
resistance

Resistance
equivalent to
mechanical load

Power relations



$$\begin{array}{ccc} P_{AG} : & P_{RCL} : & P_{conv} \\ 1 : & s : & 1-s \end{array}$$